

AI-POWERED

Document Processing & Business Workflow

AI Customer Complaint & Case Processing System

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1. Business Problem

Turning unstructured customer complaints into actionable business outputs

The challenge

Customer complaints arrive as PDFs, DOCX and text files.

Manual review is time-consuming, inconsistent and difficult to scale.

Key information is buried inside Unstructured content.

The solution

An LLM-powered workflow automatically:

- reads each document
- extracts structured case data
- drafts a customer response
- creates an internal management summary

Business value

- ✓ Faster case processing
- ✓ Consistent outputs
- ✓ Reduced manual effort
- ✓ Better case visibility
- ✓ Scalable batch processing
- ✓ Traceable structured data

2. End-to-End Workflow

A modular pipeline: ingest → understand → generate → consolidate



Design principle: keep AI tasks separate so each stage can be tested, monitored and improved independently.

3. Document Ingestion

Batch-ready file discovery and format-specific text extraction

Supported inputs

.pdf
.txt
.docx

Only eligible formats are processed.

Ingestion layer

Process_documents()

- scans the data/ folder
- identifies supported files
- extracts text
- handles read/parse errors
- logs failures without stopping the batch

Output of this stage

A clean internal document object:

Document Dataframe containing file_name, File_path, file_size and parsed_status

This becomes the input to the next AI stage.

4. Structured Information Extraction

The LLM converts unstructured complaint text into validated business data

Customer Name

Phone Number

Issue Description

Complaint: Yes/No

Supporting Document

Email

Complaint Category

Resolution Provided

Escalation Required

Overall Case Status

Pydantic schema → validation → structured JSON/object → downstream tasks

5. AI Generation Layer

Two independent outputs are generated from the same trusted case data

Customer Response Email

Purpose

Professional external communication

Includes

- appropriate greeting
- concise issue summary
- resolution / current status
- next steps when available

Internal Case Summary

Purpose

Management / operations visibility

Includes

- case overview
- key issue
- action taken
- current status
- recommended next action

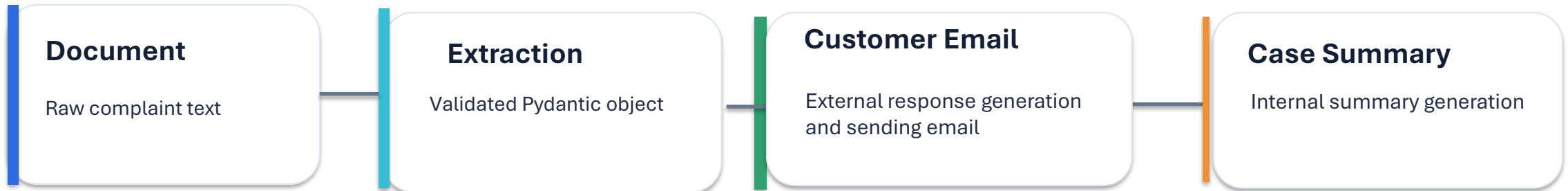
Tone

Concise, factual and operational.

Same source of truth → different audiences → different prompts

6. Sequential Workflow Design

After extraction, each task depend on results of previous task



7. Batch Processing & Error Handling

One failed document should not stop the entire batch

Batch controller

for each eligible batch:

1. ingest
2. run structured extraction
3. generate email
4. send email
5. generate summary
6. save outputs
7. update/save report

Continue processing the remaining batches.

Error handling

File errors → log + mark failed

Output errors → log + continue

Use structured logging with:
file, stage, status, error, timestamp

Operational result

Successful files produce:

- ✓ structured data csv
- ✓ customer email csv
- ✓ case summary csv
- ✓ final report csv

Failed files remain visible in the final report
with a clear processing status.

No silent failures.

8. Technology Stack

Practical GenAI engineering components

Python

Core application & orchestration

Pydantic

Schema definition and validation

LLM API

Extraction + controlled generation

Streamlit

User interface for file processing

Pandas

Final report generation

Logging

Observability and error tracking

Git / GitHub

Version control & collaboration

9. Prompt Engineering

The workflow is designed to keep AI outputs factual and structured

Extraction Process

Define the role and task clearly.

Provide the exact fields required.

Require missing information to be represented as null / unknown rather than guessed.

Return only the required structured schema.

Email Process

Use extracted case data as the source of truth.

Professional and empathetic tone.

Mention only supported facts.

If resolved, clearly communicate closure and avoid unnecessary follow-up.

Summary Process

Produce a concise internal view.

Separate facts from recommended actions.

Do not create unsupported details.

Optimize for quick management review.

PROJECT OUTCOME

From documents to decisions — automatically.

This project demonstrates a complete batch-oriented GenAI workflow with structured outputs, modular orchestration, Sequential processing, error handling and business-ready deliverables.

Ingest

Multiple file formats

Understand

Structured extraction

Generate

Email + case summary

Report

Consolidated CSV

Thank you